IL2234

Digital Systems Design and Verification using Hardware Description Languages

Project Milestone 3

Processor Design and Modelling

Introduction

In this Milestone, you will design a simple RISC-V processor. We separate the processor design in several parts for you to follow them. In any case, you must deeply understand the RISC-V instruction set architecture (ISA) with selected instructions, the memory organization, and the microarchitecture before designing one.

Instruction Set Architecture

Instruction types

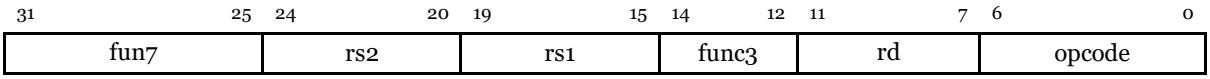
In the base ISA, there are four core instruction formats (R/I/S/U). There are two further variants of the instruction formats (SB/UJ) based on the handling of Immediate. All instructions are fixed 32 bits in length and must be aligned on a four-byte boundary in memory. Each type defines the contents and position of each element in the instruction.

Grouping instructions into types simplifies the decoding hardware, since the decoding and muxing logic needed for each instruction type can be shared across all the instructions of the same type.

R-type instructions

Register-to-register instructions. Used for **arithmetic and logical operations between two registers**; takes two source registers rs1, rs2, applies an ALU operation, and writes the result into a destination register rd.

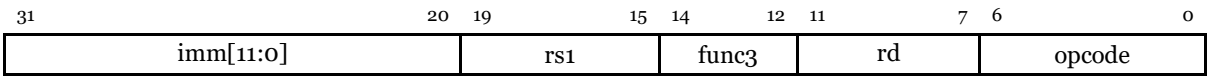
The type is described as follows:



I-type instructions

Immediate instructions. Used for **operations with immediate values, loads from memory, and jumps**; takes one source register rs1 and a 12-bit signed immediate value imm, performs the operation, and stores the result in a destination register rd.

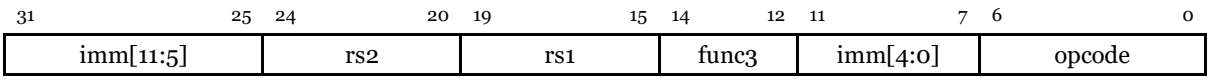
The type is described as follows:



S-type instructions

Used for **store instructions**, where data from a source register rs2 is written into memory at an address computed by adding a 12-bit signed immediate to a base register rs1; no destination register is used.

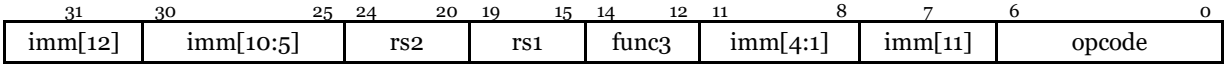
The type is described as follows:



SB-type instructions

Similar to S-type instructions, but with a different encoding of the immediate.

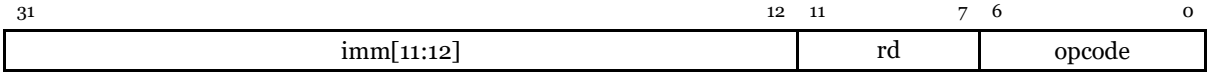
The type is described as follows:



U-type instructions

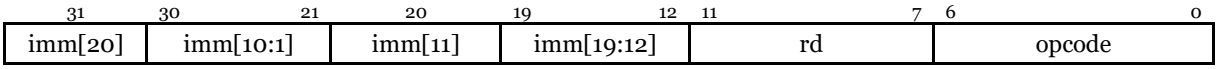
Used for **building large constants or PC-relative addressing**; loads a 20-bit immediate into the upper 20 bits of a destination register rd, with the lower 12 bits set to zero.

The type is described as follows:



UJ-type instructions

Similar to U-type instructions, but with a different encoding of the immediate.



Instruction definitions

Instructions can be grouped by functionalities, depending on what resources they use and what function they perform.

ALU instructions

ALU instructions perform arithmetic and logical operations on processor registers or immediate values.

R-type, register to register

The table below lists all the register-to-register ALU instructions. Note that the opcode is the same for all of them (0110011), and the operation is decided by the contents of func3 and func7.

Instruction	opcode	func3	func7	*Description*
ADD	0110011	000	0000000	x[rd] = x[rs1] + x[rs2]
SUB	0110011	000	0100000	x[rd] = x[rs1] - x[rs2]
SLT	0110011	010	0000000	x[rd] = (x[rs1] < x[rs2]) ? 1 : 0 (signed)
SLTU	0110011	011	0000000	x[rd] = (x[rs1] < x[rs2]) ? 1 : 0 (unsigned)
XOR	0110011	100	0000000	x[rd] = x[rs1] ^ x[rs2]
SLL	0110011	001	0000000	x[rd] = x[rs1] << x[rs2] (logical left shift)
SRL	0110011	101	0000000	x[rd] = x[rs1] >> x[rs2] (logical right shift, zero-fill)

SRA	0110011	101	0100000	x[rd] = x[rs1] >> x[rs2] (arithmetic right shift, sign-extend)
OR	0110011	110	0000000	x[rd] = x[rs1] | x[rs2]
AND	0110011	111	0000000	x[rd] = x[rs1] & x[rs2]

I-type, immediate

The table below lists all the immediate ALU instructions. Note that the opcode is the same for all of them (0010011), and the operation is decided by the contents of func3 and part of the immediate bits. Additionally, the lower 5 bits of the immediate are used to determine the shift amount shamt in the shift operations:

shamt = imm[4:0]

Note that this resembles an R-type instruction, where shamt is equivalent to rs2, and imm[11:5] to func7.

Instruction	opcode	func3	imm[11:5]	*Description*
ADDI	0010011	000	xxxxxxx	x[rd] = x[rs1] + imm
SLTI	0010011	010	xxxxxxx	x[rd] = (x[rs1] < imm) ? 1 : 0 (signed)
SLTIU	0010011	011	xxxxxxx	x[rd] = (x[rs1] < imm) ? 1 : 0 (unsigned)
XORI	0010011	100	xxxxxxx	x[rd] = x[rs1] ^ imm
ORI	0010011	110	xxxxxxx	x[rd] = x[rs1] | imm
ANDI	0010011	111	xxxxxxx	x[rd] = x[rs1] & imm
SLLI	0010011	001	0000000	x[rd] = x[rs1] << shamt (logical left shift)
SRLI	0010011	101	0000000	x[rd] = x[rs1] >> shamt (logical right shift, zero-fill)
SRAI	0010011	101	0100000	x[rd] = x[rs1] >> shamt (arithmetic right shift, sign-extend)

U-type, constant building

The table below lists the instructions used to build constants. They are not necessarily related to the ALU, but they are often used in combination with other ALU instructions to build constants.

Instruction	opcode	*Description*
LUI	0110111	x[rd] = imm[31:12] << 12
AUIPC	0010111	x[rd] = PC + (imm[31:12] << 12)

Note that these instructions do not have any func3 or func7 fields.

Memory access

Interactions for memory accesses can be divided into loads (I-type) and stores (S-type).

I-type, loads

The table below lists the load memory instructions. Note that the opcode is the same for all instructions, and the type of load is decided based on the value of func3

Instruction	*opcode*	*func3*	*Description*
LB	0000011	000	Load byte (sign-extend to 32 bits): x[rd] = signext(MEM[x[rs1]+imm][7:0])
LH	0000011	001	Load halfword (16 bits, sign-extend): x[rd] = signext (MEM[x[rs1]+imm][15:0])
LW	0000011	010	Load word (32 bits): x[rd] = MEM[x[rs1]+imm][31:0]
LBU	0000011	100	Load byte unsigned (zero-extend to 32 bits): x[rd] = MEM[x[rs1]+imm][7:0]
LHU	0000011	101	Load halfword unsigned (zero-extend): x[rd] = MEM[x[rs1]+imm][15:0]

S-type, stores

The table below lists the store memory instructions.

Instruction	*opcode*	*funct3*	*Description*
SB	0100011	000	Store **byte**: MEM[x[rs1]+imm][7:0] = x[rs2][7:0]
SH	0100011	001	Store **halfword** (16 bits): MEM[x[rs1]+imm][15:0] = x[rs2][15:0]
SW	0100011	010	Store **word** (32 bits): MEM[x[rs1]+imm][31:0] = x[rs2][31:0]

Control

Instructions are used to control the execution flow of the program by manipulating the program counter (PC). They can be further divided into three types.

SB-type, conditional branches

The table below lists the conditional branch instructions. These instructions modify the program counter based on the result of a comparison.

Instruction	*opcode*	*func3*	*Description*
BEQ	1100011	000	Branch if equal: if (x[rs1] == x[rs2]) PC += imm
BNE	1100011	001	Branch if not equal: if (x[rs1] != x[rs2]) PC += imm
BLT	1100011	100	Branch if less than (signed): if (x[rs1] < x[rs2]) PC += imm
BGE	1100011	101	Branch if greater/equal (signed): if (x[rs1] >= x[rs2]) PC += imm
BLTU	1100011	110	Branch if less than (unsigned): if (x[rs1] < x[rs2]) PC += imm
BGEU	1100011	111	Branch if greater/equal (unsigned): if (x[rs1] >= x[rs2]) PC += imm

UJ-type, unconditional jump

There is only one unconditional jump instruction: **JAL** (jump and link). This instruction updates the program counter based on an immediate value and stores the current program counter value in a specified register. This can be used for function calls, where the program has to resume execution after the function call ends.

Instruction	*opcode*	*Description*
JAL	1101111	x[rd] = PC + 4

	PC = PC + imm

Note that the program counter is stored with the increment already applied.

I-type, indirect jump

There is only one indirect jump instruction: **JALR** (jump and link register). This instruction updates the program counter based on the content of a register (and an optional immediate offset).

Instruction	*opcode*	*func3*	*Description*
JALR	1100111	000	x[rd] = PC + 4 PC = (x[rs1] + imm) & ~0b11

Note that RISC-V instructions are always 4-byte aligned, meaning valid instruction addresses must be multiples of 4. Therefore, the bitwise NOT of 0b11 (3 in decimal) ensures the lower 2 bits of the PC are forced to 0.

Others

There are other instructions in the RV32I base instruction set that are used for system calls and memory fences; these are opcodes 1110011 and 0001111. For this project, these instructions will be ignored. So, if detected, the CPU should continue with its normal execution, i.e., increment the program counter, and ignore the instruction.

For more information about the RISCV ISA, please refer to the following link: RISCV Specifications

Memory Organization

The RISC-V ISA defines 32 32-bit general-purpose registers. Additionally, it constrains the main memory by the size of the address bus and the byte-addressable requirements.

Address Bus Size

The address bus is 32 bits wide. This means the CPU can generate addresses in the range 0x00000000 – 0xFFFFFFFF.

Addressable Unit

RISC-V is a byte-addressable architecture. Each address corresponds to a single byte (8 bits).

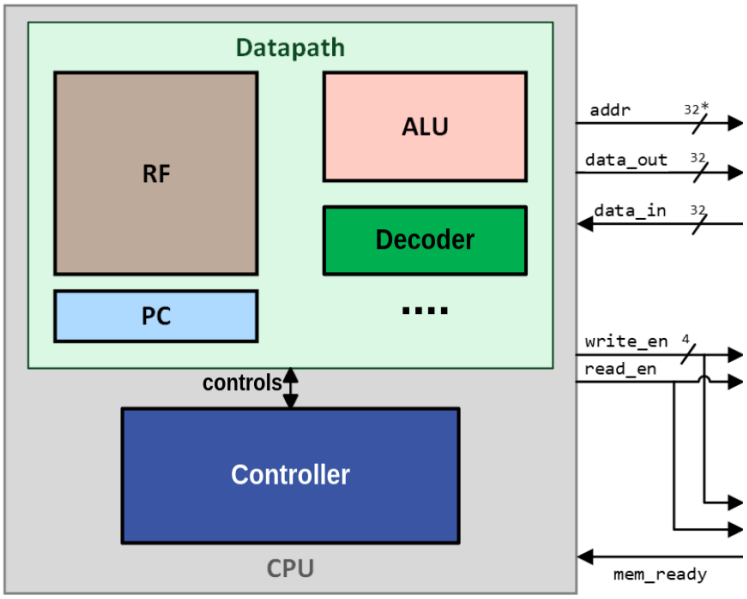
- **1 byte (8 bits):** directly accessible at any address (0,1,2,3…)
- **2 bytes (16 bits, half-word):** must be aligned to 2-byte boundaries (0,2,4,6…)
- **4 bytes (32 bits, word):** must be aligned to 4-byte boundaries (0,4,8,12…)

Addressing Modes

The addressing modes describe how memory locations are accessed. The RV32I memory access instructions define only *Register-indirect* memory accesses. This means that the accessed address is determined by the contents of one of the registers, rs1, plus an optional offset from the immediate value. There is an implicit direct memory access mode by reading from register 0, which is hardwired to zero, resulting in the immediate being the address.

Microarchitecture Design

Once the instruction set is defined, we can proceed with the microarchitectural design of datapath and controller for the CPU. The CPU has a multi-cycle controller and features a Von Neumann architecture, where the same memory is used for both data and instructions.



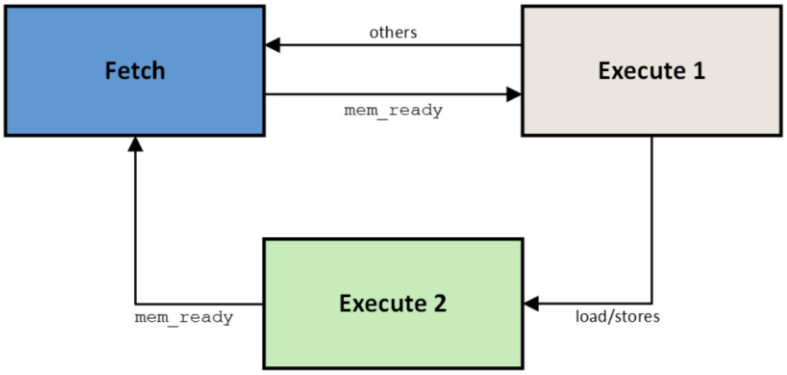
* The CPU address bus is 32 bits, but since we are using a 32-bit byte-addressable memory, the lower two bits of the address are used to generate the *write_en* signal (byte select), resulting in the "effective" address width being 30 bits. For memory behaviour, the specifications follow what you have designed in Milestone 2: Memory. Specifically, *mem_ready* indicates the completion of memory accesses both write and read.

Datapath Components

Some of the datapath components you have designed in the previous milestones, namely ALU and RF. In this milestone, you will include more logic components, such as program counter (PC), Decoder and many more. PC is used for reading the *memory instructions* from the memory, while the decoder is used to demux the *data_in* to provide data and control signals according to the ISA. You can definitely come up with your own hardware partition (not following this one).

Control State Machine

The controller will consist of 3 states: *fetch-decode*, *execute-1*, and *execute-2*.



During the *fetch-decode* state, the instruction is read from the memory. The value of program counter (PC) determines the memory address. During this state, the controller should issue the necessary control signals to enable the reading of the SRAM. The instruction is stored in the instruction register and is decoded according to the instruction types on the next rising edge of the clock.

The execution state of an instruction is determined by its type. Memory load and store operations require two execution states, while other instructions typically require just one.

During *execute-1*, the operations dictated by the instruction are carried out. For memory load and store instructions, this state computes the address (usually calculated through x[rs1]+imm). For other instruction types, the complete operation is performed within this single state. *Hint: Make most use of the ALU, particularly with zero flag. Its description is already sufficient for this simple ISA.*

During *execute-2* for loads and stores, the data is loaded/stored in the memory based on the address computed in the previous instruction.

The update of the program counter should happen at the end of the execution state. For control instructions, the PC is updated based on the results of the conditions; for the other instructions, the PC is incremented by 4. Note that an increment of 4 is needed since the instructions are 32 bits (4 bytes).

Part 1: Datapath Design

In this part, you will design and model both the datapath and control signals for a RISC-V-based processor. You will treat the ALU and Register File that you designed previously, along with additional components— like the PC and Decoder—as black boxes, but if you use simple logic components such as adder and flip flop, use appropriate illustrations for them, and then draw the complete schematic.

Start with something you already know, such as the interface of the ALU. This will help you determine where and how the components should be connected. You can then identify the remaining unknowns, such as the Decoder's output interface. At the same time, refer to the ISA and determine which components are required to execute each instruction.

Tasks

1. Draw a schematic of the CPU datapath. To do this,
   - Place all the required combinational and sequential components in the datapath, including ALU, RF, PC, decoder, all the necessary muxes, and additional logic if required.
   - Decide on the bussing structure interconnecting sequential and combinational components
   - Identify on all control signals, which will be handled by the controller, for components and buses of the datapath
2. Draw a schematic of the CPU controller, with all control signals.
3. Specify the interfaces of the unknown components: PC, decoder, and controller - including the bitwidth of each port and a description of it.

Deliverables

1. Submit an electronic drawing of your schematic in PDF format on your KTH Github repository in *report/*

Submission date: *before the examination*

Individual submission: *same content is allowed within a group*

Part 2: Controller Design

In this part, you will design the FSM state diagram for your RISC-V processor. You have already read the description of the control state machine and identified the controller interface in the previous part. You can now use this information to construct the complete FSM, determine how the control signals are assigned in each state, and define the state transitions.

Tasks

1. Draw the complete FSM based on the controller description provided in the introduction. Make sure that all control signals that are required by the interface are set correctly.

Deliverables

1. Submit an electronic drawing of your state diagram in PDF format on your KTH Github repository in *report/*

Submission date: *before the examination*

Individual submission: *same content is allowed within a group*

Part 3: SystemVerilog Modelling and Basic Verification

In this part, you will use SystemVerilog to complete the RISC-V processor. Following your schematic and FSM state diagram from the previous parts, you will implement the remaining components and integrate them into a complete processor.
You will implement the PC, Decoder, and Controller, and connect them with the previously designed components using buses, multiplexers, and additional logic as required.

Tasks

1. Implement the Decoder according to the provided RISC-V ISA.
    a. Decode the required instruction fields to identify the corresponding instruction and generate the required control and data signals based on the decoded instruction.
2. Following your datapath schematic, create an RTL design for datapath of this processor, instantiating necessary logic components and mapping them.
3. Following your FSM state diagram, implement the Controller.
   Caution: As consistently discussed in lectures, the controller merely coordinates and triggers actions by issuing appropriate control signals to the various parts of the datapath, but does not perform any computations or data operations itself.
4. Putting the datapath and controller together and create the top-level module of the processor.
5. Write a comprehensive testbench to verify all supported instructions individually. You may exclude the memory connection in this task by emulating the required memory behaviour directly in the testbench.
    a. For each instruction, verify that the correct control signals are generated.
    b. Verify that each instruction executes according to its expected behaviour as defined by the ISA.

Deliverables

1. Submit the decoder RTL design on your KTH Github repository in *riscv/decoder/*

2. Submit the controller RTL design on your KTH Github repository in *riscv/controller/*

3. Submit the top-level and the rest RTL design of the processor and its testbench(es) on your KTH Github repository in *riscv/top/*

Submission date: *before the examination*

Individual submission: *same content is allowed within a group*